

Bonus Homework 4

Discrete Mathematics by Hongfei Fu, 2023.11.23

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1. (Additional Homework)

Matroids. A finite **matroid** is an ordered pair $(E, \mathcal{I})$, where E is a finite set and $\mathcal{I} \subseteq 2^E$ is a collection of subsets of E such that

- $\emptyset \in \mathcal{I}$, and
- for any sets $A \subseteq B \subseteq E$, if $B \in \mathcal{I}$ then $A \in \mathcal{I}$, and
- for any sets $A, B \in \mathcal{I}$, if $|A| < |B|$ then there exists $x \in B \setminus A$ such that $A \cup \{x\} \in \mathcal{I}$.

Disjoint Paths. Let $G = (V, E)$ be a(n undirected) simple graph. Given a path $\rho = x_0, e_1, \dots, x_{n-1}, e_n, x_n$ in G , we denote by $\text{Edges}(\rho)$ the set of all edges appearing in ρ , i.e., $\text{Edges}(\rho) := \{e_0, \dots, e_n\}$. We say that two paths ρ, ρ' in G are **disjoint** if $\text{Edges}(\rho) \cap \text{Edges}(\rho') = \emptyset$.

Independent Sets of Vertices. Let $G = (V, E)$ be a(n undirected) connected simple graph such that $|V| \geq 2$. We assume a designated vertex v^* called the **source** vertex. A set $U \subseteq V \setminus \{v^*\}$ of vertices is called **independent** if $U = \{u_1, \dots, u_k\}$ (k can be zero) and there are k pairwise-disjoint paths $\rho_1, \dots, \rho_k$ such that each path ρ_i connects v^* and u_i (as endpoints). Define $\mathcal{I}$ to be the set of all independent sets of vertices.

Show that $(V \setminus \{v^*\}, \mathcal{I})$ is a finite matroid.

Hint: The most difficult part is to prove that $(V \setminus \{v^*\}, \mathcal{I})$ satisfies the third condition in the definition of finite matroids. Given two sets of vertices $A, B \in \mathcal{I}$, we iteratively construct a sequence of sets of vertices $C_0, C_1, \dots, C_k$ such that the final set C_k of the sequence satisfies that $A \subseteq C_k$ and $|C_k| = |B|$. Initially, we set $C_0 := B$. Then at each stage n from C_n to construct C_{n+1} , we start with choosing an arbitrary vertex $a \in A \setminus C_n$. If there is a path from v^* to a that is disjoint from all other paths that connects v^* and vertices in C_n , then we simply add the vertex a (implicitly with its path) into C_n and kick out an arbitrary vertex from $C_n \setminus A$. Otherwise, a main phenomenon is that the path ρ from v^* to a conflicts with some path ρ' from v^* to some $b \in B$. Let e be the last overlapping edge on ρ that conflicts with some ρ'' as stated above. Then we construct a path from v^* to a by first following ρ'' until e and ρ after e . By iterating the progress from C_n to C_{n+1} , the overlapping edges e can only move closer to v^* as the paths to vertices in A are guaranteed to be disjoint. Since, we cannot get infinitely closer to v^* , the construction of the sequence $C_0, C_1, \dots$ must end at some C_k , and at the end we must have that C_k is independent and $A \subseteq C_k$.

Answer Area:

- $\because \emptyset \subseteq V \setminus \{v^*\}$ and there are 0 pairwise-disjoint path.
 $\therefore \emptyset$ is **independent**.
 $\therefore \emptyset \in \mathcal{I}$.
- For $A, B, A \subseteq B \subseteq V \setminus \{v^*\}$, and $B \in \mathcal{I}$.
 $\because B \in \mathcal{I}$
 $\therefore$ There are $|B|$ pairwise-disjoint paths $\rho_1, \dots, \rho_{|B|}$ such that each path ρ_i connects v^* and u_i (as endpoints). ($u_i \in B$).
 $\because A \subseteq B$.
 $\therefore$ There are $|A|$ pairwise-disjoint paths $\rho'_1, \dots, \rho'_{|A|}$, $\{\rho'_1, \dots, \rho'_{|A|}\} \subseteq \{\rho_1, \dots, \rho_{|B|}\}$, such that each path ρ'_i connects v^* and u'_i (as endpoints). ($u'_i \in A$ and $u'_i \in B$).

$\therefore A$ is independent.

$\therefore A \in \mathcal{I}$.

- For $A, B, A, B \in \mathcal{I}$ and $|A| < |B|$.

Now we construct a sequence of sets of vertices $C_0, C_1, \dots, C_k$ such that the final set C_k of the sequence satisfies that $A \subseteq C_k$ and $|C_k| = |B|$.

Let's denote $C_0 = B$, For C_{n+1} , we start with choosing an arbitrary vertex $a \in A \setminus C_n$.

- * If there is a path from v^* to a that is disjoint from all other paths that connects v^* and vertices in C_n . We simply add the vertex a (implicitly with its path) into C_n and remove an arbitrary vertex from $C_n \setminus A$.

$$C_{n+1} = (C_n \cup a) \setminus v$$

while v is an arbitrary vertex in $C_n \setminus A$. So $|C_n|$ won't change.

- * If there isn't a path from v^* to a that is disjoint from all other paths that connects v^* and vertices in C_n .

$\therefore$ The paths from v^* to vertices in A are disjoint.

$\therefore$ The path ρ from v^* to a conflicts with some path ρ' from v^* to some $b \in C_n \setminus A$, that is, $b \in B$, since every path in $C_n \cap A$ is disjoint from ρ .

Let e be the last overlapping edge on ρ that conflicts with some ρ'' as stated above. Then we construct a path ρ^* from v^* to a by first following ρ'' until e and ρ after e . After changing the path, we can also have:

$$C_{n+1} = (C_n \cup a) \setminus v$$

while v is an arbitrary vertex in $C_n \setminus A$.

$\therefore A$ is a finite set of vertices.

$\therefore$ We can add all the vertices in A into C_n through finite steps. The construction of the sequence $C_0, C_1, \dots$ must end at some C_k , and at the end we must have that C_k is independent and $A \subseteq C_k$.

$\therefore |A| < |B|$.

$\therefore \exists x \in C_k \setminus A$, that is, $x \in B \setminus A$, the path from v^* to x is disjoint from all paths that connects v^* and vertices in A .

$\therefore$ There are $|A| + 1$ pairwise-disjoint paths $\rho_1, \dots, \rho_{|A|+1}$ such that each path ρ_i connects v^* and u_i (as endpoints). ($u_i \in A \cup \{x\}$). $\therefore$ There exists $x \in B \setminus A$ such that $A \cup \{x\} \in \mathcal{I}$.

In conclusion, $(V \setminus \{v^*\}, \mathcal{I})$ is a finite matroid.